STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA -2
Academic year	I – III Semester(2025-26)
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Branch	AI&DS
Course Outcome	4

Case Study Topic : Hyderabad Metro Emergency Routing — Bidirectional Dijkstra

A Hyderabad emergency response control room must determine the fastest route from Secunderabad Junction (SEC) to Gachibowli IT Hub (GIT) using the following 8-node Hyderabad road network:

Word Done Summary: The Hyderabad Metro Emergency Routing project was successfully developed to identify the fastest emergency travel path between metro stations during critical situations. The system models the Hyderabad Metro network as a graph, where stations are represented as nodes and routes between stations are represented as edges with associated travel distances.

To improve route-finding efficiency, the Bidirectional Dijkstra Algorithm was implemented. Unlike the traditional Dijkstra algorithm, which searches from only the source station, Bidirectional Dijkstra performs simultaneous searches from both the source and destination stations. The searches continue until they meet at an intermediate node, significantly reducing computation time and improving performance.

The project accepts a source station and destination station as input and calculates the optimal route with the minimum travel distance. It displays the emergency path, total distance, and intermediate stations involved in the route. The algorithm ensures that emergency responders and passengers can quickly identify the best route during emergencies.

Implementation Details:

Graph Modeling – Metro stations and routes were represented as a weighted graph.
Bidirectional Search – Implemented Bidirectional Dijkstra from source and destination.
Path Finding – Calculated the shortest emergency route efficiently.
Result Display – Showed the optimal path and total distance to the user.

Out put:

```
● b.deepak@BDeepaks-MacBook-Air DSA 2 % javac BidirectionalDijkstra.java
● b.deepak@BDeepaks-MacBook-Air DSA 2 % java BidirectionalDijkstra
  Shortest Distance from SEC to GIT = 33 minutes
○ b.deepak@BDeepaks-MacBook-Air DSA 2 %
```

GitHub Link:https://github.com/bdeepakg-boop/DSA-Project/upload/main

Student Signature	**Faculty Signature**
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